Solutions :

1-1. At time $N\frac{L}{R}$, the first packet has reached the destination, the second packet is stored in the last router, the third packet is stored in the next-to-last router, etc. At time $N\frac{L}{R}+\frac{L}{R}$, the second packet has reached the destination, the third packet is stored in the last router, etc. Continuing with this logic, we see that at time $N\frac{L}{R}+(P-1)\frac{L}{R}=(N+P-1)\frac{L}{R}$, all packets have reached the destination.

1-2.

a. A circuit-switched network would be well suited to the application, because the application involves long sessions with predictable smooth bandwidth requirements. Since the transmission rate is known and not bursty, bandwidth can be reserved for each application session without significant waste. In addition, the overhead costs of setting up and tearing down connections are amortized over the lengthy duration of a typical application session.

b. In the worst case, all the applications simultaneously transmit over one or more network links. However, since each link has sufficient bandwidth to handle the sum of all of the applications' data rates, no congestion (very little queuing) will occur. Given such generous link capacities, the network does not need congestion control mechanisms.

1-3.

a. Between the switch in the upper left and the switch in the upper right we can have 4 connections. Similarly we can have four connections between each of the 3 other pairs of adjacent switches. Thus, this network can support up to 16 connections.

b. We can have 4 connections passing through the switch in the upper-right-hand corner and another 4 connections passing through the switch in the lower-left-hand corner, giving a total of 8 connections.

c. Yes. For the connections between A and C, we route two connections through B and two connections through D. For the connections between B and D, we route two connections through A and two connections through C. In this manner, there are at most 4 connections passing through any link.

1-4. Tollbooths are $\frac{175}{2}=87.5$ km apart, and the cars propagate at 100 km/hour. A tollbooth services a car at a rate of one car every 12 seconds.

a. There are ten cars. It takes 120 seconds, or 2 minutes, for the first tollbooth to service the 10 cars. Each of these cars has a propagation delay of 52.5 minutes (travel 87.5 km) before arriving at the second tollbooth. Thus, all the cars are lined up before the second tollbooth after 54.5 minutes. The whole process repeats itself for traveling between the second and third tollbooths. It also takes 2 minutes for the third tollbooth to service the 10 cars. Thus the total delay is 111 minutes. The complete calculation is below:

$$
\begin{aligned}
d_{end-to-end} &= (N+1)d_{trans} + Nd_{prop} = (N+1)\frac{L}{R} + N\frac{d}{s} \\
&= 3 \times (10 \times \frac{12}{60}) + 2 \times \frac{175}{2 \times 100} \times 60 \\
&= 111 \text{ min}
\end{aligned}
$$

b. Delay between tollbooths is 8×12 seconds plus 52.5 minutes. The total delay is twice this amount plus 8×12 seconds, i.e., 109.8 minutes. The complete calculation is below:

$$
\begin{aligned}
d_{end-to-end} &= (N+1)d_{trans} + Nd_{prop} = (N+1)\frac{L}{R} + N\frac{d}{s} \\
&= 3 \times (8 \times \frac{12}{60}) + 2 \times \frac{175}{2 \times 100} \times 60 \\
&= 109.8 \text{ min}
\end{aligned}
$$

1-5.

a. $d_{prop} = \frac{m}{s}$ seconds.

b. $d_{trans} = \frac{L}{R}$ seconds.

c. $d_{end-to-end} = d_{trans} + d_{prop} = (\frac{L}{R} + \frac{m}{s})$ seconds.

d. The bit is just leaving Host A.

e. The first bit is in the link and has not reached Host B, about sd_{trans} m to arrive Host A.

f. The first bit has reached Host B.

g. From $d_{trans} = d_{prop}$ we have $\frac{L}{R} = \frac{m}{s}$, thus $m = \frac{L}{R}s = \frac{1500 \times 8}{10 \times 10^6} \times (2.5 \times 10^8) = 3 \times 10^5 \text{ m} = 300 \text{ km}.$

1-6. Consider the first bit in a packet. Before this bit can be transmitted, all of the bits in the packet must be generated. This requires $\frac{56\times 8}{64\times 10^3} = 7$ msec.

The time required to transmit the packet is $\frac{56\times 8}{10\times 10^6} = 44.8$ μsec.

Propagation delay = 10 msec.

The delay until decoding is $7\text{m} + 44.8\mu + 10\text{m} = 17.0448\text{m}$ sec.

1-7. The first end system requires $\frac{L}{R_1}$ to transmit the packet onto the first link; the packet propagates over the first link in $\frac{d_1}{s_1}$; the packet switch adds a processing delay of d_{proc}; after receiving the entire packet, the packet switch connecting the first and the second link requires $\frac{L}{R_2}$ to transmit the packet onto the second link; the packet propagates over the second link in $\frac{d_2}{s_2}$. Similarly, we can find the delay caused by the second switch and the third link: $\frac{L}{R_3}$, d_{proc} , and $\frac{d_3}{s_3}$.

Adding these five delays gives $d_{end-to-end} = \sum_{i=1}^{3}(\frac{L}{R_i} + \frac{d_i}{s_i}) + 2d_{prop}$.

To answer the second question, we simply plug the values into the equation to get $4.8 + 4.8 + 4.8 + 20 + 16 + 4 + 3 + 3 = 60.4$ msec.

1-8. Because bits are immediately transmitted, the packet switch does not introduce any delay; in particular, it does not introduce a transmission delay. Thus, $d_{end-to-end} = \frac{L}{R} + \sum_{i=1}^{3}\frac{d_i}{s_i}$

For the values in Problem 10, we get $4.8 + 20 + 16 + 4 = 44.8$ msec.